

Background

- Cystic fibrosis transmembrane conductance regulator (CFTR) modulators have revolutionized cystic fibrosis (CF) treatment, but adherence remains critical for their effectiveness¹
- CFTR modulators for people with certain CFTR mutations include ivacaftor, lumacaftor/ivacaftor, tezacaftor/ivacaftor, and elexacaftor/tezacaftor/ivacaftor
- While adherence to CFTR modulators is generally expected to be higher due to their oral administration and effectiveness, there is limited data available²
- Pharmacy Quality Alliance recognizes adherence standard as a PDC threshold of 80%³
- Non-adherence in CF can result in hospitalizations, poor health outcomes, and significant personal and economic impacts⁴
- This study aims to retrospectively evaluate procedure modifications to observe if a change in adherence to CFTR modulators in a specialty pharmacy setting occurred

Objectives

Primary Objective

Change in proportion of days covered (PDC) of nonadherent patients on CFTR modulators who utilized Saturday delivery between May-October 2024

Secondary Objectives

- Medication refill method in nonadherent patients who utilized Saturday delivery
- Percentage of overall utilization of Saturday delivery in patients on CFTR modulators
- Patient reported outcomes in nonadherent patients post intervention

Methods

Study Design

- Maxor Specialty Pharmacy services patients nationally
- A longitudinal retrospective chart review gathered data from descriptive reports from 2 locations to assess for inclusion from November 1, 2023, to October 31, 2024

Data Collection & Analysis

- The pharmacy operating system and supporting clinical platforms provided information for data collection
- Collected data included age, zip code, first administration date, medication dose, frequency, duration, PDC, day of week refilled, and patient reported outcomes
- PDC was calculated as days with medication available divided by number of days from index start thru last day of observation period with a 6 month lookback
- PDC defined as very adherent >90%, adherent 80-89%, nonadherent <80%
- Data was analyzed using descriptive statistics

Inclusion Criteria

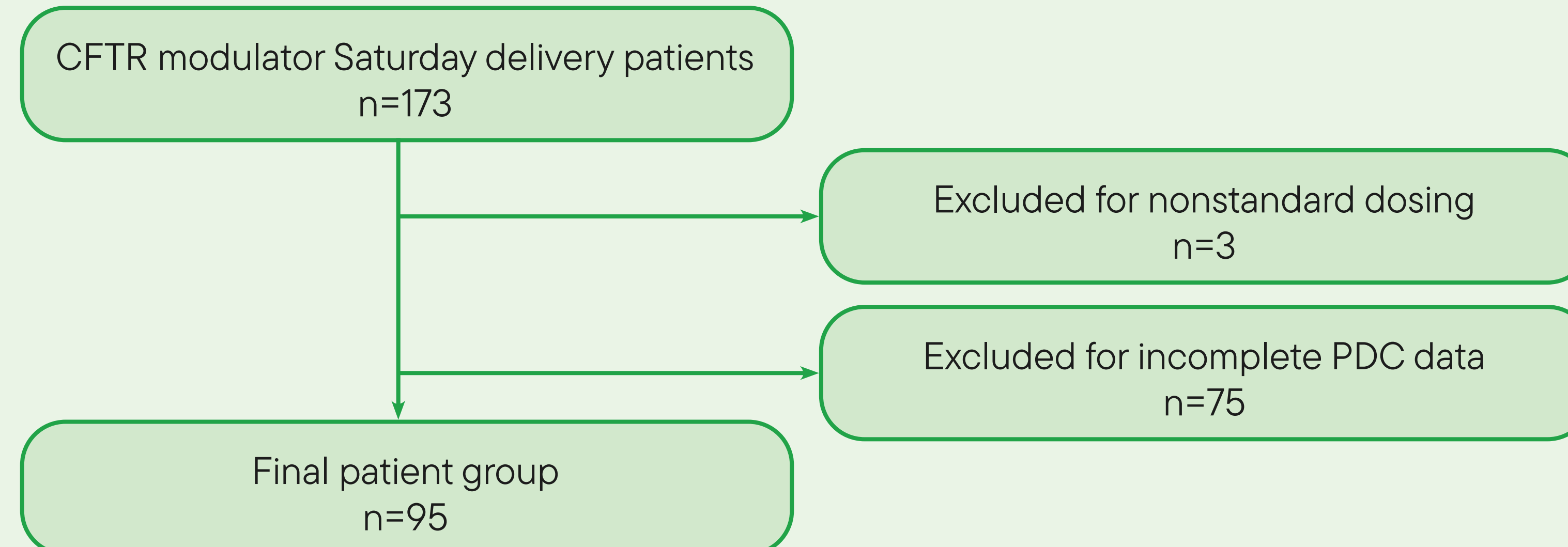
Patients with CF who filled a CFTR modulator prescription through Maxor Specialty Pharmacy prior to November 1, 2023, utilized Saturday delivery since May 1, 2024, and had complete PDC data through October 31, 2024

Exclusion Criteria

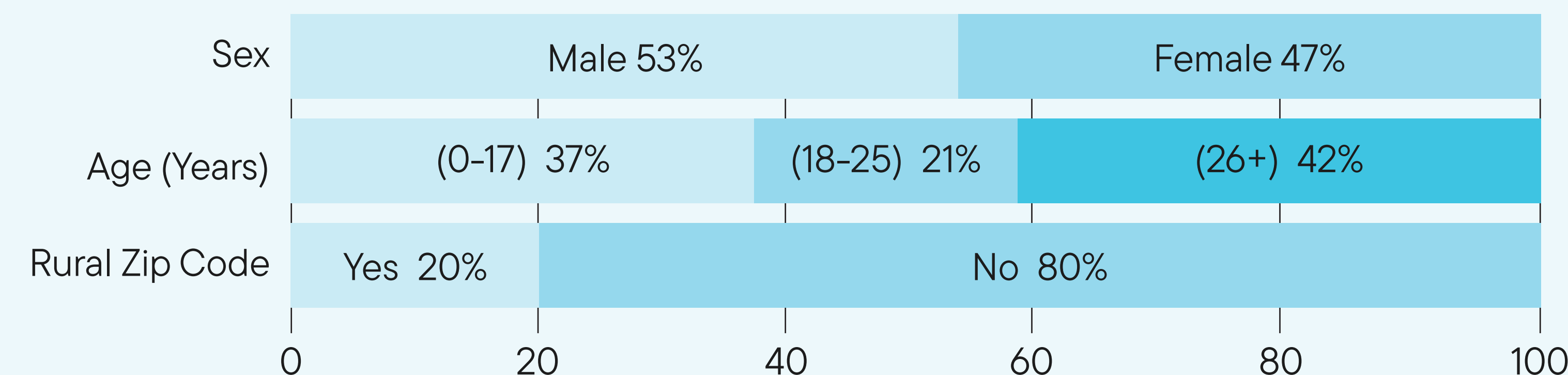
Patients who do not have CF, patients with CF not on a CFTR modulator, patients who received their initial CFTR modulator order after November 1, 2023, patients with incomplete PDC data, and patients on nonstandard dosing

Results

Eligibility Criteria



Patient Demographics



Primary Objective

Nonadherent Patients n=21		52% of nonadherent patients (n=11) moved to adherent Of the newly adherent patients, – 36% (n=4) increased to very adherent – 45% (n=5) had >20% increase in PDC
Pre Saturday Delivery (PDC Average)	Post Saturday Delivery (PDC Average)	
64.9%	71.1%	

Very adherent: >90% PDC; Adherent: 80-89% PDC; Nonadherent: <80% PDC

Secondary Objectives

Nonadherent Patient Refill Method			3% utilization (n=185) of Saturday delivery for CFTR modulator prescriptions
	Pre Saturday Delivery	Post Saturday Delivery	
Outbound Call	68.9%	55.4%	
Inbound Call	28.9%	33.9%	8.5%
Click to Refill*	2.2%	10.7%	

*Digital engagement sent via text message

Nonadherent Patient Reported Outcomes			7%
	Pre Saturday Delivery	Post Saturday Delivery	
Remained Stable	95.9%	91.3%	
Gotten Worse	4.1%	1.7%	7%
Improved	0%	7%	

Discussion

Conclusions

- 22.1% of patients were identified as nonadherent
- Transition of care aged patients between 18-25 years old were 8% more likely to be nonadherent
- 43% of nonadherent patients had a rural zip code compared to 20% of the overall patient population
- There was a 6.2% increase in PDC averages for nonadherent patients after Saturday delivery was offered
- Greater than 50% of the previously nonadherent patients achieved adherence after Saturday delivery implementation
- Staff phone efficiency with nonadherent patients increased after intervention as there was a 13.5% decrease in outbound refill calls and 8.5% increase in click to refill
- Overall utilization of Saturday delivery for CFTR modulator prescriptions was 3% with 12 patients utilizing Saturday delivery more than once
- Nonadherent patient reported outcomes improved by 7% and decreased worsening by 2.4%

Limitations

- Small sample size due to limited study patient eligibility from incomplete PDC data
- Given the small sample size, the evaluation was not powered to provide statistical significance, so descriptive statistics were evaluated only
- PDC cannot account for intentional gaps in therapy and can overestimate nonadherence due to administrative delays
- Because PDC is a claims based metric, refill data can indicate potential adherence but does not verify actual usage
- Other interventions to improve compliance ran concurrently with the Saturday delivery intervention, making it difficult to determine a direct causal relationship
- Other noncontrollable factors that potentially influenced the data include seasonal adherence changes, increase in alternative dispensing organizations, reenrollment requirement changes, payer delays, and technology outages

References

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